# Phase 34: Finite Partial Functions

Finite partial functions are partial functions with a finite domain. The operator $\twoheadrightarrow$ denotes a finite partial function, meaning the function may not be defined everywhere, and its domain must be finite.

## Example 1 : Basic Finite Function Type

A finite partial function from X to Y is a partial function whose domain is finite. This is useful when modeling resources with limited capacity or databases with a bounded number of entries.

$$[Year, TableFF]$$

$$\begin{array}{|l} records : Year \twoheadrightarrow TableFF \\ \hline \#(\mathrm{dom}\, records) \leq 1000 \end{array}$$

The records relation is a finite partial function because: 1. It is partial: not all years need to have a table 2. Its domain is finite: at most 1000 years have records 3. It is functional: each year maps to at most one table

## Example 2 : Comparison with Other Function Types

Comparison of function type operators:

- Total function ($\longrightarrow$): defined for all inputs, domain is entire source set - Partial function ($\pfun$): may not be defined for all inputs, domain may be infinite - Finite partial function ($\twoheadrightarrow$): partial function with finite domain - Total bijection ($\rightarrowtail\!\!\!\!\!\twoheadrightarrow$): defined everywhere, injective and surjective

$$[X, Y]$$

$$\begin{array}{|l} totalFunc : X \longrightarrow Y \\ partialFunc : X \pfun Y \\ finitePartialFunc : X \twoheadrightarrow Y \\ totalBij : X \rightarrowtail\!\!\!\!\!\twoheadrightarrow Y \\ \hline \mathrm{dom}\, totalFunc = X \land \mathrm{dom}\, totalBij = X \land (\forall x1, x2 : X \mid totalBij(x1) = totalBij(x2) \bullet x1 = x2) \land \mathrm{ran}\, totalBij = \end{array}$$

## Example 3 : Finite Functions elem Practice

Real-world example: A database table storing customer preferences.

$$[CustomerID, Preference]$$

$$\begin{array}{|l} preferences : CustomerID \twoheadrightarrow Preference \\ \hline \#(\mathrm{dom}\, preferences) \leq 10000 \end{array}$$

This models a preference database where: - Not all customers have preferences recorded (partial) - The database has a capacity limit (finite domain) - Each customer maps to one preference value (functional)

## Example 4 : Operations on Finite Functions

$$
\begin{array}{|l}
f : \mathbb{N} \pfun \mathbb{N} \\
g : \mathbb{N} \pfun \mathbb{N} \\
\hline
f = \{1 \mapsto 10, 2 \mapsto 20, 3 \mapsto 30\} \land \\
\quad g = \{10 \mapsto 100, 20 \mapsto 200\} \land \\
\quad \#(\mathrm{dom}\, f) = 3 \land \\
\quad \#(\mathrm{dom}\, g) = 2
\end{array}
$$

We can compose finite functions, but the result may not be finite unless proven.

$$
\begin{array}{|l}
f\_composed : \mathbb{N} \pfun \mathbb{N} \\
\hline
f\_composed = g \fatsemi f \land \#(\mathrm{dom}\, f\_composed) \leq \#(\mathrm{dom}\, f)
\end{array}
$$

The composition has a domain size bounded by the domain of f, since we can only compose where f's range intersects g's domain.

## Example 5 : Domain Restrictions land Finiteness

$[Title, Length, ViewDate]$

$$
\begin{array}{|l}
viewed : Title \pfun ViewDate \\
recent\_viewed : Title \pfun ViewDate \\
recentSet : \mathbb{P}\ Title \\
\hline
recentSet = \{\, t : Title \mid t \in \mathrm{dom}\ viewed \,\} \land \\
\quad recent\_viewed = recentSet \lhd viewed \land \\
\quad \#(\mathrm{dom}\ recent\_viewed) \leq \#(\mathrm{dom}\ viewed)
\end{array}
$$

Restricting a finite partial function preserves finiteness. The recent_viewed function has a domain that is a subset of viewed's domain, so it remains finite.